

Loan Repayment Risk Prediction Using Machine Learning

An Imbalance-Aware and Interpretable Approach

PROJECT TEAM

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PREDICTION TASK

Repayment difficulty

EVALUATION DESIGN

Imbalance-aware

EVIDENCE STANDARD

Untouched test set

Predicting repayment difficulty from applicant and credit history

Binary prediction: will an applicant have repayment difficulty (TARGET)? The classes are severely imbalanced.

INPUTS

Application and historical records

- Current application information
- Historical credit records
- Historical repayment records

History is brought together at applicant level.

OUTPUTS AND GOALS

A reproducible, interpretable pipeline

The model estimates repayment difficulty from the available records.

Predictive outputs do not establish causality or guarantee repayment outcomes.

Seven Home Credit tables, one row per applicant

SOURCE TABLE	ROWS	COLUMNS
application_train	307,511	122
bureau	1,716,428	17
bureau_balance	27,299,925	3
previous_application	1,670,214	37
POS_CASH_balance	10,001,358	8
installments_payments	13,605,401	8
credit_card_balance	3,840,312	23

RAW CSVS

About 2.3 GB

AGGREGATION KEY

SK_ID_CURR

MODEL INPUTS

134 features

History is aggregated to SK_ID_CURR, giving one model row per applicant.

A leakage-safe pipeline from raw tables to demo

01

Validate raw data

7 tables and data
validation

02

Aggregate history

Applicant-level
aggregation

03

Engineer features

Financial and
repayment feature
engineering

04

Split applicants

Stratified: 60% train
(184,506), 20% validation
(61,502), 20% test (61,503);
seed 42

05

Fit preprocessing

Leakage-safe
preprocessing, fitted on
training only

06

Compare and tune

Baseline comparison,
then controlled
XGBoost tuning

07

Select and freeze

Validation-based
threshold selection,
frozen final model

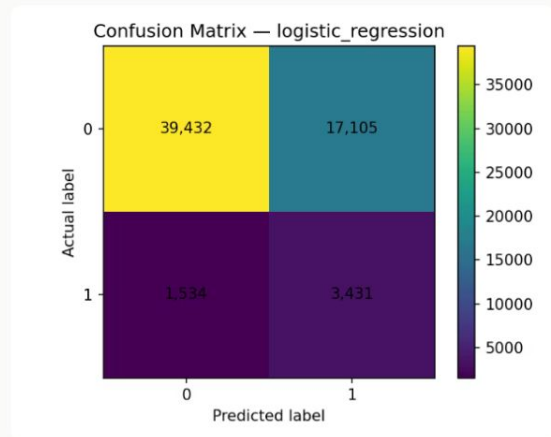
08

Evaluate and explain

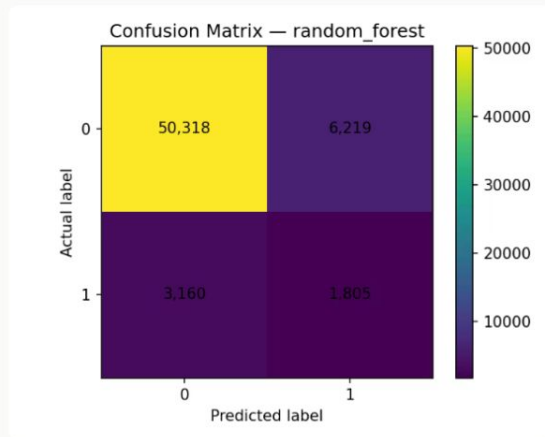
Untouched test
evaluation, permutation
importance, Streamlit
demo

Validation confusion matrices for the three baselines

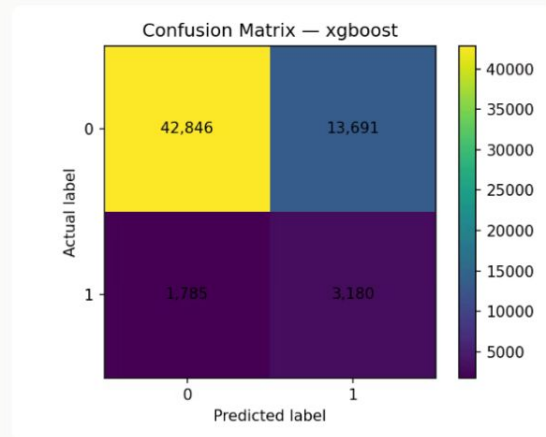
Validation / model-comparison outputs, not final test results.



VALIDATION — LOGISTIC REGRESSION



VALIDATION — RANDOM FOREST



VALIDATION — XGBOOST

Supplied figures are shown unchanged. These diagnostics belong to baseline comparison, not the frozen model's final test evaluation.

Baseline Model Comparison

Best AUC performance

XGBoost	
ROC-AUC	0.7722
PR-AUC	0.2591
Accuracy	0.7484
Precision	0.1885
Recall	0.6405
F1	0.2913
Training time	17.3 s

Highest recall, slowest training

Logistic Regression	
ROC-AUC	0.7599
PR-AUC	0.2366
Accuracy	0.6969
Precision	0.1671
Recall	0.6910
F1	0.2691
Training time	555.0 s

Highest accuracy

Random Forest	
ROC-AUC	0.7409
PR-AUC	0.2076
Accuracy	0.8475
Precision	0.2250
Recall	0.3635
F1	0.2779
Training time	68.2 s



TAKEAWAY

XGBoost had the strongest ROC-AUC and PR-AUC and was selected as the candidate for controlled tuning.

Baseline vs Tuned XGBoost

Controlled hyperparameter tuning gave a modest improvement in validation performance at the same threshold (0.50).

Baseline XGBoost

Validation • Threshold 0.50

ROC-AUC	0.7722
PR-AUC	0.2591
Accuracy	0.7484
Precision	0.1885
Recall	0.6405
F1	0.2913



Tuned XGBoost

Validation • Threshold 0.50

ROC-AUC	0.7745
PR-AUC	0.2616
Accuracy	0.7869
Precision	0.2073
Recall	0.5807
F1	0.3056



Change (Tuned – Baseline)

Validation • Threshold 0.50

Metric	Δ (Tuned – Baseline)	Direction
ROC-AUC	+0.0023	▲
PR-AUC	+0.0026	▲
Accuracy	+0.0386	▲
Precision	+0.0188	▲
Recall	-0.0598	▼
F1	+0.0143	▲



KEY TAKEAWAY

Controlled tuning produced a modest improvement in validation PR-AUC, ROC-AUC and F1, **while recall decreased**. All results are on the validation set at the same threshold (0.50).

Tuned XGBoost: Threshold 0.50 vs 0.58

A higher decision threshold (0.58) improves F1 by trading recall for higher precision, while the AUCs remain unchanged.

Tuned XGBoost

Validation • Threshold 0.50

ROC-AUC	0.7745
PR-AUC	0.2616
Accuracy	0.7869
Precision	0.2073
Recall	0.5807
F1	0.3056



Tuned XGBoost

Validation • Threshold 0.58

ROC-AUC	0.7781
PR-AUC	0.2738
Accuracy	0.8466
Precision	0.2561
Recall	0.4729
F1	0.3322



Change (0.58 – 0.50)

Validation

Metric	Δ (0.58 – 0.50)	Direction
ROC-AUC	+0.0036	▲
PR-AUC	+0.0122	▲
Accuracy	+0.0597	▲
Precision	+0.0488	▲
Recall	-0.1078	▼
F1	+0.0266	▲



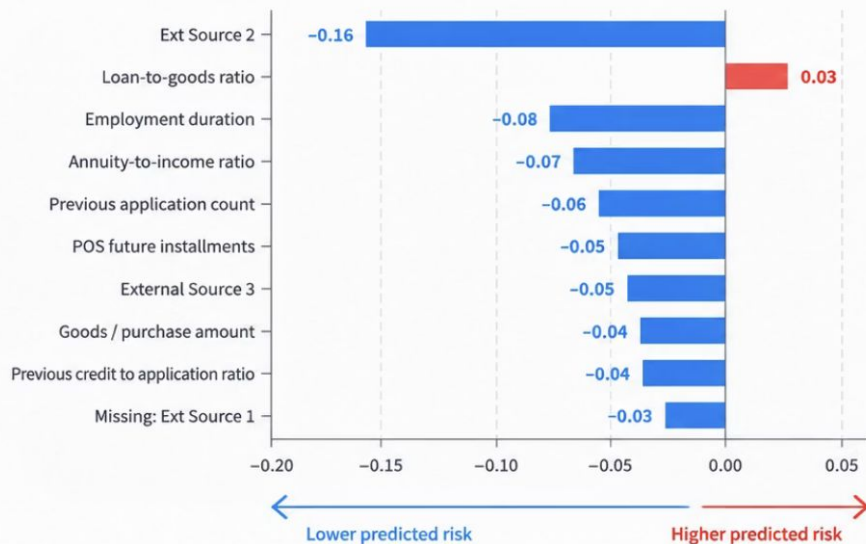
KEY TAKEAWAY

Increasing the decision threshold from **0.50** to **0.58** improves precision (0.2073 → 0.2561) and overall F1 (0.3056 → 0.3322), while reducing recall (0.5807 → 0.4729). Accuracy also increases (0.7869 → 0.8466).

ROC-AUC (0.7745 → 0.7781) and PR-AUC (0.2616 → 0.2738) remain largely unchanged, as thresholding does not affect the underlying probability ranking.

Interpretability And Streamlit Demo

The chart shows the strongest individual model contributions for this applicant. Positive contributions push the model toward repayment difficulty; negative contributions push it toward lower risk.



Historical feature reference

Neutral historical reference

Only the application details below are entered by the user. Historical bureau/payment variables are supplied by the selected synthetic reference profile. For the academic demo, use the high-risk stress-test reference with the deliberately adverse applicant values to demonstrate a high-risk decision.

Age (years)

35.00 - +

Employment duration (years)

5.00 - +

Number of children

1 - +

Family members

3.00 - +

Annual income (₹)

300000 - +

Annual repayment / annuity (₹)

25000 - +

Loan / credit amount (₹)

500000 - +

Goods / purchase price (₹)

450000 - +

ASSESS REPAYMENT RISK

Final Results & Key Takeaways

ROC-AUC

0.7781

Measures overall class discrimination.

PR-AUC

0.2738

Especially informative for the imbalanced positive class.

Precision

0.2561

25.61% of predicted high-risk cases were correct.

Recall

0.4729

47.29% of actual repayment-difficulty cases were identified.

F1 Score

0.3322

Balances precision and recall.



Final Tuned XGBoost

Frozen threshold = 0.58 | Evaluated once on the untouched test set.